Manish Kumar

manishkumarmaurya8873@gmail.com | +91-8873679268 | LinkedIn | GitHub | Portfolio

Professional Summary

Associate Software Engineer at Accenture working on ServiceNow ticket triage and routing. Built real-time voice applications and tools for IT operators, including a spoken English tutor with synchronized avatar playback and an incident agent that retrieves runbooks before proposing a fix.

Technical Skills

Languages: Python, TypeScript, JavaScript, C/C++, SQL, HTML, CSS
AI & Retrieval: Multi-agent systems, retrieval-augmented generation (RAG), ChromaDB, Azure OpenAI embeddings, Groq, tool calling
Machine Learning & Spatial Data: scikit-learn, pandas, anomaly detection, PostGIS, Folium
Speech & Audio: Deepgram Nova-3 (speech recognition), Aura (speech synthesis), OpenSpline avatar animation, Web Audio API, AudioWorklet
Backend & Data: FastAPI, WebSockets, asyncio, Firebase Cloud Firestore, Vercel Serverless Functions
Web & Mobile: React, React Native, Expo, Tailwind CSS, Streamlit, UPI deep links
Deployment & Tools: Docker, Render, Vercel, Expo EAS Build, Git, GitHub, Linux

Experience

Associate Software Engineer, Accenture Aug 2026 – Present

- Develop backend components for ServiceNow workflows that categorize incoming tickets and route them to the appropriate team.

Software Developer Intern, Accenture Dec 2025 – May 2026

- Integrated AI-assisted triage into IT service management workflows to help operators review tickets and escalate incidents.

Projects

Speak with Sway – Live AI English Tutor [Live] [GitHub]

- Built a spoken English tutor that gives phrasing feedback and asks follow-up questions using Deepgram speech recognition and synthesis, Groq, and FastAPI.
- Integrated OpenSpline to synchronize the tutor's facial movements with its voice. Learners can interrupt a reply and continue the conversation.
- Streamed audio over WebSockets with a bounded buffer to keep playback smooth and preserve packet order.

Autonomous Infrastructure Agent – Incident Diagnosis [Live] [GitHub]

- Connected a diagnostic agent to incident runbooks using ChromaDB and Azure OpenAI embeddings. A planning agent turns the diagnosis into proposed tool calls.
- Built a Streamlit dashboard where operators can simulate incidents, review a proposed fix, approve a simulated action, and inspect the audit log.

Apna Kirana – Mobile Ordering & Store POS [Live Demo] [GitHub]

- Built a commission-free hyperlocal ordering platform with a React Native (Expo) customer app and a real-time web POS for store counters.
- Used Firestore subscriptions to update the store counter as orders arrive. Added Hindi voice announcements and UPI deep links with the merchant and order total prefilled.

Vessel Anomaly Detection – Maritime Machine Learning [GitHub]

- Flagged unusual vessel movements in AIS position reports using Isolation Forest, Local Outlier Factor, and an autoencoder. Preserved raw coordinates for PostGIS queries and interactive Folium maps.

Education

Rajiv Gandhi Institute of Petroleum Technology 2022 – 2026
B.Tech. in Computer Science and Engineering CPI: 7.69
Dr. I.S.N.R.K.P.L.D. Kisan College, Sitamarhi – Intermediate: 80.8% 2019 – 2021
Simultala Awasiya Vidyalaya – Matriculation: 94.2% 2018 – 2019

Achievements

- **Winner**, IIT Mandi Hackathon | **Grand Finalist**, IIT Bombay Hackathon
- **JEE Advanced Top 2%** | **Bihar Board State Rank 13** in Matriculation | **NTSE Stage 1 Qualified**